

```
1  module seq_counter #(
2      parameter CLK_FREQ      = 50_000_000,
3      parameter UPDATE_HZ     = 2,
4      parameter DISP_WIDTH    = 10
5  )(
6      input                clk_i,
7      input                rst_ni,
8
9      output reg [2:0]      column_o,
10     output reg            row_o
11 );
12
13 localparam CNT_STOP = CLK_FREQ / UPDATE_HZ;
14 localparam WIDTH = $clog2(CLK_FREQ);
15
16 reg [WIDTH-1:0] cnt_o;
17 wire            tick_w = (cnt_o == CNT_STOP - 1);
18
19 always @(posedge clk_i or negedge rst_ni)
20     if (!rst_ni) cnt_o <= 0;
21     else if (tick_w) cnt_o <= 0;
22     else cnt_o <= cnt_o + 1;
23
24
25 always @(posedge clk_i or negedge rst_ni)
26     if (!rst_ni) row_o <= 1'b0;
27     else if (tick_w)
28         if (row_o == 1'b0 && column_o == 3'b101) row_o <= 1'b1;
29         else if (row_o == 1'b1 && column_o == 3'b000) row_o <= 1'b0;
30
31
32 always @(posedge clk_i or negedge rst_ni)
33     if (!rst_ni) column_o <= 3'd0;
34     else if (tick_w)
35         if (row_o == 1'b0 && column_o != 3'b101) column_o <= column_o + 1'b1;
36         else if (row_o == 1'b1 && column_o != 3'b000) column_o <= column_o - 1'b1;
37
38 endmodule
```